

MCQs:-

1. C
2. A
3. D
4. B
5. C
6. A
7. D
8. B
9. C
10. A

Q:2

Atmosphere:- The envelope of gases & water vapour surrounding the planet Earth is known as Atmosphere.

Pollutant:- Anything that is in air, water or soil which has a harmful effect on some part of the environment is known pollutant.

Percentage of nitrogen:- 78%.

Percentage of oxygen:- 21%.

Percentage of water vapour:- 4%.

Q:3.

- ① Carbonmonoxide $\{CO\}$
- ② ~~At~~ Nitrogen oxide $\{NO, NO_2\}$
- ③ Ground level ozone $\{O_3\}$
- ④ Methane $\{CH_4\}$
- ⑤ Sulphur oxides $\{SO_2, SO_3\}$
- ⑥ Carbondioxide $\{CO_2\}$

* Sulphur oxides: SO_2 , SO_3

source: - power stations and industries using fossil fuels.

harmful effect: - lung cancer, acid rain etc.

* Nitrogen Oxides: NO , NO_2

sources: - Exhaust fumes of motor vehicles etc.

harmful effect: - coughs, headaches.

Q:4

(i) Carbon monoxide is produced by the incomplete combustion of fuels. A person cannot detect it because it is colourless and odorless.

(ii) CO combines strongly with haemoglobin to form carboxyhaemoglobin. This reduces the blood's ability to carry oxygen ~~to~~ around the body.

(iii) Ground level ozone is called a secondary pollutant because it is emitted directly. It is formed in the atmosphere when nitrogen oxides & hydrocarbons react in sunlight.

Q.5

(i) 4 air pollutants.

Pollutant	One source	One harmful effect.
Carbon monoxide	Incomplete burning of woods & fuels.	headache, Dizziness, death & damage.
Sulphur dioxide	Power stations & industries using fossil fuels.	lung cancer, bronchitis & acid rain.
Nitrogen oxide	Exhaust fumes of motor vehicles.	Cough, headaches, lung disease.
Methane	Wet lands, rice fields, Landfills	Global warming.

(ii)

Incomplete combustion occurs when a fuel burns with an insufficient supply of O_2 .

It produces carbon monoxide (CO).

(iii)

In a closed room, the oxygen supply becomes limited as the heater burns. This causes more incomplete combustion & producing carbon monoxide, which builds up and makes the room increasingly dangerous.

Precaution : Keep a window open to provide ventilation.

Q: 6

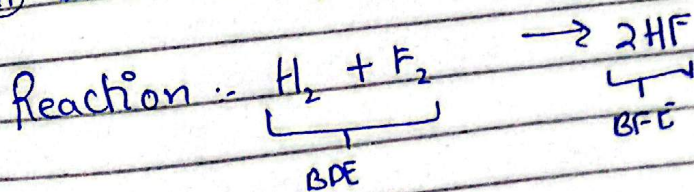
(i) Enthalpy of reaction : The heat energy change when a reaction takes place at constant pressure.

* Exothermic ΔH is negative (-).

* Endothermic ΔH is positive (+).

(ii)
$$\Delta H^\circ = \frac{\text{Sum of bond dissociation energies of reactants}}{\text{Sum of bond formation energies of products}}$$

(iii) $\Delta H^\circ = \text{sum of BDE} - \text{sum of BFE}$.



formula : $\text{sum of bond dissociation energies of reactants} - \text{sum of bond formation energies of products}$.

$$\Delta H^\circ = [\text{BDE of H-H} + \text{BDE of F-F}] - (2 \times \text{BFE of H-F})$$

$$\Delta H^\circ = [436 \text{ kJ/mol} + 155 \text{ kJ/mol}] - (2 \times 567 \text{ kJ/mol})$$

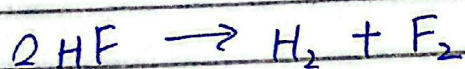
$$\Delta H^\circ = 591 \text{ kJ/mol} - 1134 \text{ kJ/mol}$$

$$\Delta H^\circ = -543 \text{ kJ/mol}$$

(iv)

The reaction is exothermic bcz its ΔH° is negative (-543 kJ/mol) meaning energy is released.

(v) for the reverse reaction :-



$$\Delta H^\circ (2 \times 567) - (436 + 155)$$

$$\Delta H^\circ = 1134 \text{ kJ/mol} - 591 \text{ kJ/mol}$$

$$\Delta H^\circ = +543 \text{ kJ/mol}$$

It is positive bcz the reverse reaction absorbs the same amount of energy that the forward reaction releases.